

SOLUTION REQUIREMENTS

Strategic Product Placement Analysis

Purpose

The solution requirements define the functional, data, visualization, web integration, and quality needs of the Strategic Product Placement Analysis project. These requirements ensure that retail data is analyzed clearly and presented through an interactive dashboard, analytical story, and web interface.

FUNCTIONAL REQUIREMENTS

- Analyze average sales volume across product categories.
- Compare product performance across aisle, end-cap, and front-of-store positions.
- Analyze consumer demographic groups and their sales patterns.
- Study the relationship between foot traffic and average sales volume.
- Compare product price with competitor price.
- Review promotional and seasonal sales patterns.
- Provide interactive filters for exploring dashboard data.
- Present important findings through a Tableau story.

DATA REQUIREMENTS

- Retail data must contain relevant product and sales attributes.
- Required fields should support category, position, sales volume, demographics, foot traffic, pricing, promotion, and seasonal analysis.
- Data values should be checked for missing, inconsistent, or unsuitable records before visualization.
- Field names and data types should be suitable for Tableau analysis.

VISUALIZATION REQUIREMENTS

- Charts must be readable, consistent, and directly related to business questions.
- Dashboard filters must allow users to explore important retail dimensions.
- Titles, labels, legends, and tooltips should clearly describe the visualized measures.
- The dashboard should present multiple insights in a clean layout.
- The Tableau story should organize key findings in a logical sequence.

WEB INTEGRATION REQUIREMENTS

- The project must use Flask to serve the web application.
- The frontend should use Bootstrap-based responsive styling.
- The Tableau dashboard must be embedded in the website.
- The Tableau story must remain interactive after embedding.
- Navigation should provide direct access to Home, About, Dashboard, Story, and Project Information sections.

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QUALITY AND SYSTEM REQUIREMENTS

Requirements for a Clear and Reliable Project Experience

Requirement Area	Requirement	Expected Result
Usability	The interface and visualizations should be simple to understand and navigate.	Users can explore the project without unnecessary complexity.
Performance	Embedded Tableau views and Flask pages should load correctly under normal use.	Dashboard and story content remain accessible and responsive.
Compatibility	The web application should work in a modern desktop web browser.	Project sections and embedded visualizations display correctly.
Interactivity	Dashboard filters and story navigation should remain functional.	Users can actively explore visual insights.
Maintainability	Project files should follow a clear folder structure with separate templates and static assets.	The Flask project is easier to manage and update.
Presentation	The interface should use consistent headings, spacing, and project-specific content.	The final project appears clean and professional.

Final Solution Requirement

The final solution must provide an interactive retail analytics experience that combines Tableau-based product placement analysis with a Flask and Bootstrap web interface.

The solution should help users compare sales volume, product categories, store positions, customer demographics, foot traffic, pricing, promotions, and seasonal patterns. The dashboard and story must present these insights clearly and remain interactive when embedded in the web application.

Requirement Analysis Outcome

The identified requirements provide a clear foundation for project design and development. They connect the retail analysis goals with the required data, Tableau visualizations, interactive features, and Flask web integration. These requirements guide the project toward a simple, structured, and useful product placement analysis solution.

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